# SimpleGrid problems compiled to d-DNNF with d4 (projected, no evidence): means by grid size

| Grid | Problems | Horizon | d4 time (s) | Total time (s) | CNF variables | CNF clauses | Projected variables | Circuit nodes | Circuit edges | Model count | Query with 2 obs. (s) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3×3 | 4 | 5 | 0.04 | 2.0 | 141 | 818 | 98 | 44 | 84 | 18 | 0.001 |
| 4×4 | 4 | 7 | 0.04 | 1.9 | 400 | 2,609 | 291 | 228 | 451 | 140 | 0.003 |
| 5×5 | 4 | 8.5 | 0.09 | 2.0 | 812 | 5,621 | 604 | 1,108 | 2,210 | 1,017 | 0.006 |
| 6×6 | 4 | 10.75 | 1.36 | 4.0 | 1,557 | 11,191 | 1,175 | 21,140 | 42,296 | 24,171 | 0.017 |
| 7×7 | 4 | 12 | 6.78 | 11.7 | 2,436 | 17,936 | 1,854 | 82,748 | 165,584 | 88,126 | 0.036 |

Each row is the arithmetic mean over that grid size's problems (Problems gives how many), rounded; the per-problem values are in compile-d4.md. Each problem's evidence-free recognition theory — the problem at its recognition horizon, with the goal literals as the hypotheses, as recognize.sh builds it — compiled by marginals.sh `--solver` d4 `--project`. The projection keeps the drive actions and the final-slice goal atoms, and FiFO first verifies that they determine every other atom, so the projected circuit gives exact counts; any pinned observations are unit literals over the kept actions, so one saved circuit answers every evidence file by conditioning, without recompiling. CNF variables and clauses are those of the ground theory d4 is given; projected variables are the ones it keeps. d4 time is the d4 run alone; total time is the whole marginals.sh run (reading the theory, the projection check, d4, importing and saving the circuit, and the marginals of the unconditioned circuit). Circuit nodes and edges are those of d4's decision-DNNF, before FiFO smooths it. Times are wall-clock, 1 compilation at a time, each limited to 30 minutes. SBCL ran with an 8 GB heap and a 1 GB control stack, d4 with a 64 MB stack (the defaults, 1 GB, 2 MB and 8 MB, are too small for the larger circuits). The circuits were imported by FiFO with its d4 importer rewritten to use bit-vector scopes and iterative traversal (branch ddnnf-import-bitvector-scopes); the released importer runs out of memory on circuits of about a million nodes. Model count (verified) is the number of models of the compiled circuit (all literals set to 1 in the smoothed circuit), that is, the number of truck walks the theory allows. Every circuit kept was checked against an independent count of those walks by dynamic programming over the grid (H-1 steps, each a drive to a neighbouring cell or a wait, from p1-1 to a goal cell), which also agrees with sharpSAT on every problem shown; the two counts were equal for every circuit in the table. Grids larger than 7×7 are omitted: from 8×8 on, d4 often crashes (bus error or segmentation fault) or returns a wrong circuit without any error, and repeated runs on the same CNF give different circuits. Query with 2 obs. is the time to recompute every marginal of the saved circuit conditioned on two observations — the first two drives of a shortest plan to the problem's alphabetically first destination, pinned to steps 1 and 2 (evidence/PROBLEM-DESTINATION-obs2.txt) — as marginals.sh `--circuit --evidence-file` does: the observations are unit literals over kept atoms, so the circuit is clamped and re-evaluated, not recompiled. It is timed inside SBCL, the median of 5 repetitions, and excludes starting SBCL and loading FiFO (about 1.2 s) and reading the circuit file (0.001–0.45 s). Each answer was checked: both observed drives have probability 1, and the final-slice goal marginals sum to 1. FiFO evaluates the circuit in two linear passes; its backward pass through an AND node of k children uses prefix and suffix products (an earlier version recomputed each child's k-1 sibling products, $k^2$ multiplications, and took up to 7 s on 7×7 — smoothed AND nodes here have up to 1,007 children).